

Perturbation based validation of GSDME dependence in silico

DDR ON in all conditions | asynchronous fixed-point endpoints

Reference: DDR ON, GSDME endogenous	2%	0%	93%	0%	4%
GSDME E1	100%	0%	89%	0%	5%
GSDME KO	0%	5%	91%	0%	4%
CASP3 E1	100%	0%	93%	0%	0%
GSDME KO + CASP3 E1	0%	100%	93%	0%	0%
CASP3 KO	0%	0%	91%	0%	4%
miR-204-5p E1	49%	0%	0%	0%	51%
miR-204-5p E1 + GSDME KO	0%	48%	0%	0%	52%
miR-204-5p E1 + CASP3 KO	0%	0%	0%	0%	41%
SIRT1 KO	96%	0%	0%	0%	4%
SIRT1 KO + GSDME KO	0%	91%	0%	0%	9%
MALAT1 KO	5%	0%	86%	0%	9%
MALAT1 KO + GSDME KO	0%	10%	85%	0%	5%
GSDME E1 + CASP3 E1	100%	0%	88%	0%	0%
SIRT1 KO + GSDME KO + CASP3 E1	0%	100%	0%	0%	0%
p53 E1 + miR-204-5p E1 + CASP3 KO + p21 E1	0%	0%	0%	0%	100%
MALAT1 E1 + miR-204-5p KO + SIRT1 E1	0%	0%	100%	0%	0%
BAX E1	8%	0%	91%	0%	1%
BAX E1 + GSDME KO	0%	9%	90%	0%	1%

Output ON
100%
75%
50%
25%
0%

GSDME-mediated pyroptosisApoptosisResistanceProliferationCell-cycle arrest

Outputs may coexist. GSDME E1 clamps DFNA5 ON; it does not measure GSDME cleavage.